



Ear, nose and throat involvement in granulomatosis with polyangiitis: how it presents and how it determines disease severity and long-term outcomes

Mara Felicetti¹ · Diego Cazzador² · Roberto Padoan¹ · Alfonso Luca Pendolino² · Chiara Faccioli² · Ennio Nardello² · Alvise Berti³ · Marina Silvestrini⁴ · Giuseppe Paolazzi³ · Giuliano Brunori⁵ · Elisabetta Zanoletti² · Enzo Emanuelli² · Alessandro Martini² · Franco Schiavon¹

Received: 22 December 2017 / Revised: 27 January 2018 / Accepted: 31 January 2018 / Published online: 20 February 2018
© International League of Associations for Rheumatology (ILAR) 2018

Abstract

Ear, nose and throat (ENT) manifestations in granulomatosis with polyangiitis (GPA) represent the most frequent symptoms at disease onset. The aim of the study was to analyse ENT involvement at diagnosis, as well as how it could influence relapse rate, mortality and disease severity. A retrospective non-controlled cohort study was performed including all consecutive diagnosed GPA from 1996 to 2016 in two rheumatology centres of Northern Italy, focusing particularly on ENT presenting signs and symptoms at baseline. Eighty-nine patients (48.3% females) with new onset GPA were evaluated. They were mostly Caucasian (97.7%), middle aged (mean 54.5 years) and more frequently anti-neutrophil cytoplasmic antibodies (ANCA) positive (78.6%) with PR3 specificity (81.4%). At diagnosis, ENT involvement was reported in 71.9% patients, second only to systemic symptoms. These patients were significantly younger at disease onset (0.013), with less frequent renal involvement (0.014) irrespectively to ANCA status, but with significantly higher Vasculitis Damage Index (VDI) (0.001). The most frequent ENT manifestation was sinonasal involvement (58.4%, 73% of which with nasal inflammation/chronic sinusitis and 48% with nasal crusting), while otologic involvement (mainly otitis media/otomastoiditis) was observed in 34.8%. ENT-GPA patients presented a higher survival rate at 5 years (98.1 vs 77.7%, 0.049), and ENT involvement resulted to be an independent predictor of better outcome (OR 0.37, 95% CI 0.2–0.8, 0.019). Our data confirms that ENT involvement is not only one of the key clinical features of GPA, but also could point out a milder GPA subset with lower renal involvement and lower mortality rate, irrespectively to ANCA status.

Keywords Granulomatosis · Polyangiitis · Long-term outcomes · ENT involvement · Damage · ANCA

Introduction

Granulomatosis with polyangiitis (GPA), previously known as Wegener's granulomatosis [1], is a systemic autoimmune

disease of unknown aetiology characterised by necrotising granulomatous inflammation and vasculitis affecting mainly small blood vessels [2]. The prevalence of GPA ranges between 3 cases per 100,000 inhabitants in the USA [3] and 16 cases per 100,000 inhabitants in southern Sweden [4].

Clinical manifestations of GPA can be heterogeneous, affecting the upper respiratory tract, lungs and kidneys, often at the same time. The collaborative clinical trials conducted by the European Vasculitis Study (EUVAS) group demonstrated that patients could be classified in different levels of disease severity, ranging from a localised to an early systemic, a generalised, a severe and finally a refractory form, depending on type, location and severity of clinical manifestations with different treatment protocols [5]. In the generalised form, anti-neutrophil cytoplasmic antibodies (ANCA) direct against proteinase 3 (PR3) are found in 70–80% of patients, whereas ANCA against

✉ Roberto Padoan
robertopadoan.reumatologia@gmail.com

¹ Operative Unit of Rheumatology, Department of Medicine DIMED, University of Padova, Via Giustiniani, 2, 35128 Padova, Italy
² Otorhinolaryngology Unit, Department of Neurosciences, University of Padova, Padova, Italy
³ Rheumatology Unit, Santa Chiara Hospital, Trento, Italy
⁴ Otolaryngology Unit, Santa Chiara Hospital, Trento, Italy
⁵ Nephrology Unit, Santa Chiara Hospital, Trento, Italy

myeloperoxidase (MPO) occurs in only 10% of the patients. Localised forms of the disease in the head and neck region are not exceptional [6, 7], and ANCA could be recognised in only 60% of cases [8]. The prevalence of this involvement in GPA patients differs in literature, mainly because of lack of specificity regarding ear, nose and throat (ENT) symptoms. A retrospective study on 53 patients in Nottingham, UK, reported an ENT involvement in 80% of patients [9]; a Mayo Clinic review on 411 cases found ENT manifestations in 72.3% [10], while in a recent prospective study it was 87% [11].

However, the diagnosis of GPA, especially in the localised form, remains challenging thus contributing to the delay in the beginning of treatment. This could lead to important sequelae in the head and neck area, due to tissue destruction [12].

The aim of our study was to evaluate the prevalence of head and neck manifestations at disease onset in our cohort of GPA patients, with particular emphasis on their presentation forms, in order to contribute to early diagnosis.

Moreover, since it has also been recently reported that ENT involvement is associated with milder forms of GPA and an increased mortality risk in patients without ENT symptoms [13, 14], we analysed the differences in terms of relapse rate, mortality and disease severity in our GPA cohort, according to presence or absence of ENT involvement.

Methods

Patient selection

The study included all newly diagnosed GPA patients, between 1996 and 2016 in two different vasculitis regional referral centres of Northern Italy, the Department of Rheumatology of the University of Padova and the Santa Chiara Hospital in Trento (Operative Unit of Rheumatology and the Operative Unit of Nephrology) with at least one laryngologist evaluation at diagnosis. All GPA diagnoses were revised by physicians expert on vasculitis, according to the 2007 EMA algorithm [15]. All patients were evaluated by an otorhinolaryngologist expert on vasculitis of the Otorhinolaryngology Unit of Padova University and the Otorhinolaryngology Unit of Santa Chiara Hospital in Trento.

Data collection

The clinical charts of all included patients were retrospectively reviewed, through independent evaluations performed, by a senior expert team with a uniform and homogeneous policy in each centre. Data collected at the first interview included demographic and clinical features at diagnosis (baseline), organ involvement, laboratory findings (erythrocyte sedimentation rate (ESR), C-reactive protein (CRP), serum creatinine, presence of urinary casts, proteinuria 1+, hematuria and 24-h

urinary albumin secretion above 0.5 g) and indexes for disease activity and damage, respectively Birmingham Vasculitis Activity Score version 3 (BVASv3) [16] and Vasculitis Damage Index (VDI) [17]. For each patient, estimated glomerular filtration rate (eGFR) at baseline was calculated applying CKD-EPI 2009 formula [18]. Significant renal impairment was defined with a newly observed GFR below 60 ml/min or, when lacking these data, if the medical charts reported history of acute kidney injury (AKI) [19] or rapidly progressive renal failure.

At the last follow-up, we collected overall survival, major relapse rate and free major relapse survival after first remission induction. Major relapses were defined when presenting at least a major item of BVASv3 score or when the disease could not be managed only with steroid dosage increase.

ENT assessment

ENT clinical evaluation at baseline consisted in a nasal endoscopic examination with high-definition rigid endoscopes to ascertain the presence of nasal crusting, nasal inflammation, bloody rhinorrhoea and sinusitis. Otological examination combined with pure tone audiometry, tympanometry and acoustic reflex testing was performed to evaluate possible conductive or sensorineural hearing loss. Finally, an endoscopic laryngoscope was used to investigate the status of the pharynx and larynx. If required, radiological imaging by means of a head and neck or temporal bone computed tomography (CT) and/or head and neck magnetic resonance imaging (MRI) was performed as further investigation, in order to assess any radiological sign of organ damage (i.e. extension of septal perforation, bone erosion).

Statistical analysis

According to the ENT involvement at diagnosis, two patient cohorts were identified in the series (ENT-GPA and non-ENT-GPA). Categorical variables are presented as frequency and percentages, while continuous data are reported as median and interquartile range or mean $\pm$ standard deviation (SD), as appropriate. The chi-square test or the Fischer exact test was used for categorical variables, and the Mann–Whitney test or the *t* test for independent samples was applied to continuous variables.

Long-term outcomes considered in our analysis were relapse rate, number of relapse/year and survival rate at 12–24 and 60 months of follow-up. The estimated distributions of major relapse of the disease and overall survival were performed with the Kaplan–Meier method and the log-rank test. Cox regression test was applied to identify the predictors of outcomes, using as covariate those with significant *p* values at univariate analysis. A *p* value < 0.05 was considered significant.

All statistical analyses were carried out with IBM SPSS Statistics for Windows, Version 24.0, IBM Corp., Armonk, NY (released in 2016), and GraphPad Prism version 5.00 for Windows, GraphPad Software, San Diego, California, USA.

Results

At baseline

After excluding 19 GPA patients due to missing ENT status at diagnosis (no clinical or radiological data available), 89 patients were assessed, almost all Caucasians (97.7%), with no significant sex differences (43 females, 48.3%, and 46 males, 51.7%). Seventy patients (78.6%) were ANCA positive, 57 (64%) with PR3 specificity and 13 (14.6%) with MPO specificity, while the remnant 19 patients (21.4%) were ANCA negative. No double-positive ANCA specificity (PR3-MPO) was recorded in our cohort. The main clinical and demographic features at disease onset, according to ENT involvement, are listed in Table 1.

GPA patients presented a median age of 54.5 years at disease onset (range 14–83 years). The patients' cohort with ENT involvement was significantly younger than the group without ENT manifestations, with median age of 51 years (range 44–60) vs 61 years (range 52–68), respectively ($p = 0.013$). The diagnostic latency was found to be higher in ENT patients, but without reaching a significant difference. Interestingly, no significant differences in ANCA status were observed between ENT-GPA and non-ENT-GPA patients.

The diagnosis was histologically confirmed in 55 (67.9%) of the patients. Among these patients, 26 (47.3%) specimens were taken from the ENT area. In the other cases, biopsy was performed on the kidney, lung, nerve or skin.

ENT involvement was observed in 64 patients (71.9%) and resulted to be the most frequent GPA organ-specific presenting manifestation in our cohort, after only to systemic symptoms, reported in 66 patients (74.1%). Lung involvement was observed in 67.4% of the patients, kidney in 62.9%, nervous system in 33.7%, eye in 25.8% and skin in 21.3%. Gastrointestinal tract involvement was observed in only two (2.2%) patients, while no cardiovascular symptoms were reported at baseline. The comparison of the two groups did not show a significant difference in organ involvement at baseline, except for renal manifestations, which were more frequent and severe in non-ENT-GPA patients [21 (84%) vs 35 (54.7%), OR = 4.4; 95% CI = 1.3–14.1; $p = 0.014$]. In addition, in this group of patients, a significantly higher serum creatinine [2.1 (0.9–3.8) vs 1.0 (0.7–1.8), $p = 0.025$], a lower eGFR (32 vs 75 ml/min, $p = 0.015$) and a higher rate of AKI [16 (64.0%) vs 19 (30.6%), (OR = 4.0; 95% CI = 1.5–10.7; $p = 0.007$) were observed. Despite this, no significant differences of BVASv3 were noted between the two groups, while a VDI above 0 at

baseline was more frequently associated with ENT involvement [24 (37%) vs 1 (4%), $p = 0.001$].

ENT manifestations

The systematic evaluation of GPA patients in an ENT clinic with rigid or flexible endoscopes and pure tone audiometry were essential to the establishment of disease activity and damage.

The frequencies of ENT manifestations at diagnosis are reported in Table 2. The most frequently observed ENT manifestation in GPA patients at diagnosis was the sinonasal involvement (58.4%). Nasal inflammation (42.7%), chronic sinusitis (42.7%) and nasal crusting with or without bloody rhinorrhoea (28.1%) were the most common symptoms in patients with active disease, as shown in Fig. 1. In case of highly suspicious lesions, nasal mucosa biopsy was performed under local anaesthesia, with negligible side effects rate. Eighteen patients (20.3%) already presented a cartilage or bone damage at onset; in particular, 11 patients (12.4%) had evidence of bone erosion on the CT scan and in 7 cases (7.9%) septal perforation was observed (of these, in 2 patients it was so extensive as to lead to a nasal bridge collapse), as shown in Fig. 2.

Disease activity in the ear manifested as a combination of hearing loss associated with otalgia and inflammation of the tympanic membrane or the surrounding tissues. Otitis media/otomastoiditis was the most frequent otologic manifestations (22.5%); otitis externa the less frequent one. Conductive hearing loss was observed in 17 patients, and 11 patients had an asymmetrical sensorineural hearing loss.

Only one case of hypoglottic stenosis was found in a 47-year-old male patient presenting with hoarseness, stridor and acute respiratory distress, but without any nasal manifestations. Direct bronchoscopy examination revealed a narrowing in the subglottic region craniocaudally extended for about 10 mm.

Long-term outcomes

At the last follow-up (median 54.5 months, range 1–228 months), we registered seven deaths, all in ANCA-positive patients. Demographic and clinical features with the cause of death of each patient are listed in Table 3. Overall survival rate resulted to be 97.7% at 12 months, 96.5% at 24 months and 92.9% at 60 months (Fig. 3a).

Survival rate at 5 years resulted to be significantly worse in elderly patients (age ≥ 65 years, 75.8 vs 93.6%, increased mortality rate 17.8%, $p = 0.002$) and in pANCA/MPO-positive patients (62.7 vs 98.1%, increased mortality rate 35.4%, $p < 0.001$). On the other hand, ENT-GPA patients presented a significantly higher survival rate at 5 years (98.1 vs 77.7%, decreased mortality 20.4%, $p = 0.04$) (Fig. 3b–d). Cox regression analysis on survival data was performed only in

Table 1 Clinical and demographic features at disease onset, according to ENT involvement

	<i>GPA</i> (<i>n</i> 89)	<i>ENT-GPA</i> (<i>n</i> 64)	<i>Non-ENT-GPA</i> (<i>n</i> 25)	<i>p</i>
Age at disease onset (years)	54.5 (44–65)	51 (44–60)	61 (52–68)	0.013
Female, <i>n</i> (%)	43 (48)	32 (50)	11 (44)	ns
Caucasian, <i>n</i> (%)	87 (97.8)	63 (98.4)	24 (96)	ns
ANCA by IF or ELISA, <i>n</i> (%)				
pANCA/MPO	13 (15)	9 (14)	4 (16)	ns
cANCA/PR3	57 (64)	39 (61)	18 (72)	
Negative	19 (21)	16 (25)	3 (12)	
Diagnostic latency (months)	4 (2–7)	4 (2–6)	4 (2–8)	ns
AAV-positive biopsy, <i>n</i> (%)	55 (62)	43 (74)	12 (52)	0.056
Serum creatinine (mg/dl)	1.1 (0.8–2.3)	1.0 (0.7–1.8)	2.1 (0.9–3.8)	0.025
eGFR (ml/min)	68 (29–96)	75 (36–100)	32 (14–87)	0.015
CRP (mg/l)	43.8 (16–149)	56.8 (18–149)	35.3 (12.4–154)	ns
ESR (mm/h)	68.5 (52–96)	70 (52–98)	67 (49–84)	ns
Organ involvement, <i>n</i> (%)				
Systemic	66 (74.1)	40 (64.5)	16 (64)	ns
Lung	60 (67.4)	43 (68.3)	17 (68)	ns
Kidney	56 (62.9)	35 (54.7)	21 (84)	0.014
Acute kidney injury	35 (39.3)	19 (29.6)	16 (64)	0.007
CNS/PNS	30 (33.7)	20 (31.3)	10 (40)	ns
Eye	23 (25.8)	18 (28.1)	5 (20)	ns
Skin	19 (21.3)	14 (21.9)	5 (20)	ns
BVASv3 (median ± SD)	19.3 ± 8.6	19.5 ± 9.1	18.7 ± 7.3	ns
VDI, <i>n</i> (%)				
= 0	64 (72)	40 (63)	24 (96)	0.001
> 0	25 (28)	24 (37)	1 (4)	

Data are expressed as median and interquartile range

GPA granulomatosis with polyangiitis, *ENT* ear, nose and throat, *n* number, *ANCA* anti-neutrophil cytoplasmic antibody, *IF* immunofluorescence microscopy, *ELISA* Enzyme-Linked Immuno Assay, *PR3* anti-proteinase-3 antibody, *MPO* anti-myeloperoxidase antibody, *AAV* ANCA-associated vasculitis, *eGFR* estimated glomerular filtration rate, *CRP* C-reactive protein, *ESR* erythrocyte sedimentation rate, *PNS/CNS* peripheral nervous system/central nervous system, *BVASv3* Birmingham Vasculitis Activity Score (version 3), *SD* standard deviation, *VDI* Vasculitis Damage Index, *ns* not significant

Table 2 Frequency of ENT manifestation at diagnosis

	<i>GPA patients</i> , <i>n</i> (%)
Sinonasal	52 (58.4)
Mucosal inflammation	38 (42.7)
Chronic sinusitis	38 (42.7)
Nasal crusting/bloody rhinorrhoea	25 (28.1)
Bone erosion	11 (12.4)
Septal perforation/nasal bridge collapse	7 (7.9)
Otologic	31 (34.8)
Otitis media/otomastoiditis	20 (22.5)
Conductive hearing loss	17 (19.1)
Asymmetrical sensorineural hearing loss	11 (12.4)
Perichondritis	5 (5.6)
Otitis externa	4 (4.5)
Laryngeal	
Hypoglottic stenosis	1 (1.1)

GPA granulomatosis with polyangiitis, *ENT* ear, nose and throat, *n* number

ANCA-positive patients with age, ANCA status and ENT involvement as covariates due to the low numbers of events and no deaths observed in the ANCA-negative group. The pANCA/MPO specificity resulted to be an independent

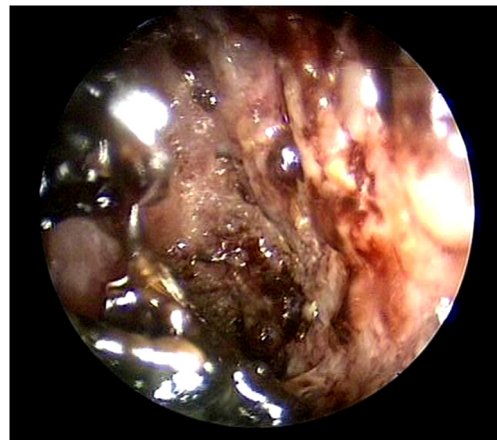


Fig. 1 Left nasal cavity, endonasal view with a 30° endoscope showing diffuse crusting and infection. Endonasal anatomical structures are indistinguishable

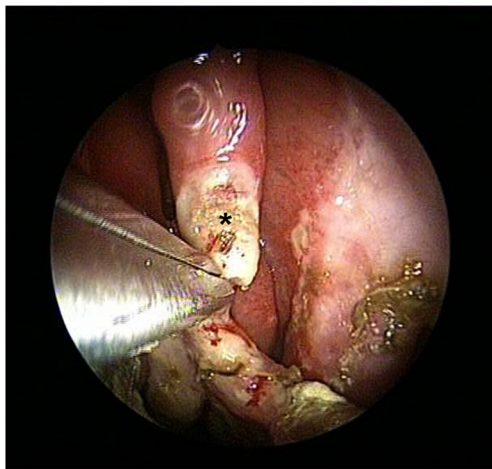


Fig. 2 Anterior septal perforation with active mucosal ulcerations at the margins. A mucosal biopsy with Weil-Blakesley forceps is performed under a 0° endoscopic view at the posterior margin of the lesion, where vomer bone erosion is evident (*)

predictor of mortality (OR 4.5, 95% CI 1.9–10.7, $p < 0.001$), while ENT involvement was significantly associated with a better outcome (OR 0.37, 95% CI 0.2–0.8, $p = 0.019$).

The overall major relapse rate increased progressively during the follow-up from 16.9% in the first year up to 41.2% at 5 years (23.8% at 2 years) (Fig. 4a). The relapse rate and the number of major relapses/year were significantly higher in young patients (49.0 vs 10.8%, $p = 0.009$) while a longer remission maintenance was observed in patients with absent or milder renal involvement (defined as $\text{GFR} \geq 60 \text{ ml/min}$) (relapse rate 48.5 vs 20.4%, $p = 0.04$) (Fig. 4b, c). Interestingly, the relapse rate or the number of relapses/year did not differ between the ANCA status (negative, cANCA/PR3 and pANCA/MPO) and in the presence of ENT involvement. Cox regression analysis on major relapse identified that an age above 65 years resulted in protective against disease relapses (OR 0.5, 95% CI 0.3–0.9, $p = 0.03$).

Discussion

Up to now, GPA diagnosis remains challenging, with a high rate of diagnostic delay in the absence of typical signs or symptoms. Since ENT involvement often represents the first manifestation of the disease, the otorhinolaryngologist could play a key role in GPA management [20]. Presence or absence of ENT involvement is an important clinical parameter for disease severity class [5, 21].

In our cohort, ENT manifestations were frequent, second only to systemic symptoms, and represents the most frequent GPA organ-specific presenting condition. In particular, we found that the nasal area was the most frequently affected by the disease (58.4%), with a high prevalence of inflammatory mucosal lesions, chronic rhinosinusitis, nasal crusting and bloody rhinorrhoea. Our results are in line with those of other authors. Hoffman and colleagues found an ENT involvement at diagnosis in 73% of GPA patients [22], while in other previous case series, nasal symptoms were reported in 55–90% [9, 11, 23, 24].

Saddle nose deformity and/or nasal septal perforation were consistent with the prospective data offered by Del Pero and collaborators who analysed 138 GPA patients and finally reported 13% of cartilaginous/bony damages over a follow-up period of 60 months [11]. This might have led to a higher Vasculitis Damage Index (VDI) in the ENT-GPA patients, which then correlates with the damage in the ENT area.

Otological manifestations generally occur between 20 and 70% of cases, with a prevalent involvement of the middle ear (40–70%), leading to a higher frequency of conductive hearing loss than the sensorineural one, as it was seen in our cohort [25, 26]. Hypoglossic stenosis is rare (2–20%) and mainly associated with a systemic involvement of GPA, as observed in one only patient of our series [11, 23, 27–29].

Table 3 Demographic and clinical features with cause of death of the deceased patients

Patient	Sex	Age at onset	ANCA	BVASv3 at onset	EUVAS severity	FFS5	Organ involvement at onset			Survival months (years)	Cause of death	BVASv3 at LFU	VDI at LFU
							ENT	AH	GFR < 60 ml/min				
1	F	54	MPO	5	Limited	0	Present	Absent	Absent	79 (7)	Infection	0	7
2	M	68	MPO	18	Severe	2	Absent	Absent	Present	30 (3)	Unknown	NA	5
3	F	83	MPO	12	Generalised	2	Absent	Absent	Present	58 (5)	Infection	0	2
4	M	60	PR3	9	Limited	0	Present	Absent	Absent	21 (2)	Cancer	0	3
5	F	81	MPO	18	Severe	2	Absent	Present	Present	3 (0)	AH	NA	NA
6	M	83	MPO	21	Generalised	2	Absent	Absent	Present	7 (0)	HF	12	NA
7	F	71	MPO	18	Generalised	1	Present	Absent	Absent	101 (8)	Stroke	0	6

Only patient 6 had known previous cardiac disorder

ANCA anti-neutrophil cytoplasmic antibodies, BVASv3 Birmingham Vasculitis Activity Score version 3, VDI Vasculitis Damage Index, ENT ear, nose and throat, AH alveolar haemorrhage, GFR glomerular filtration rate, LFU last follow-up, NA not available, HF heart failure

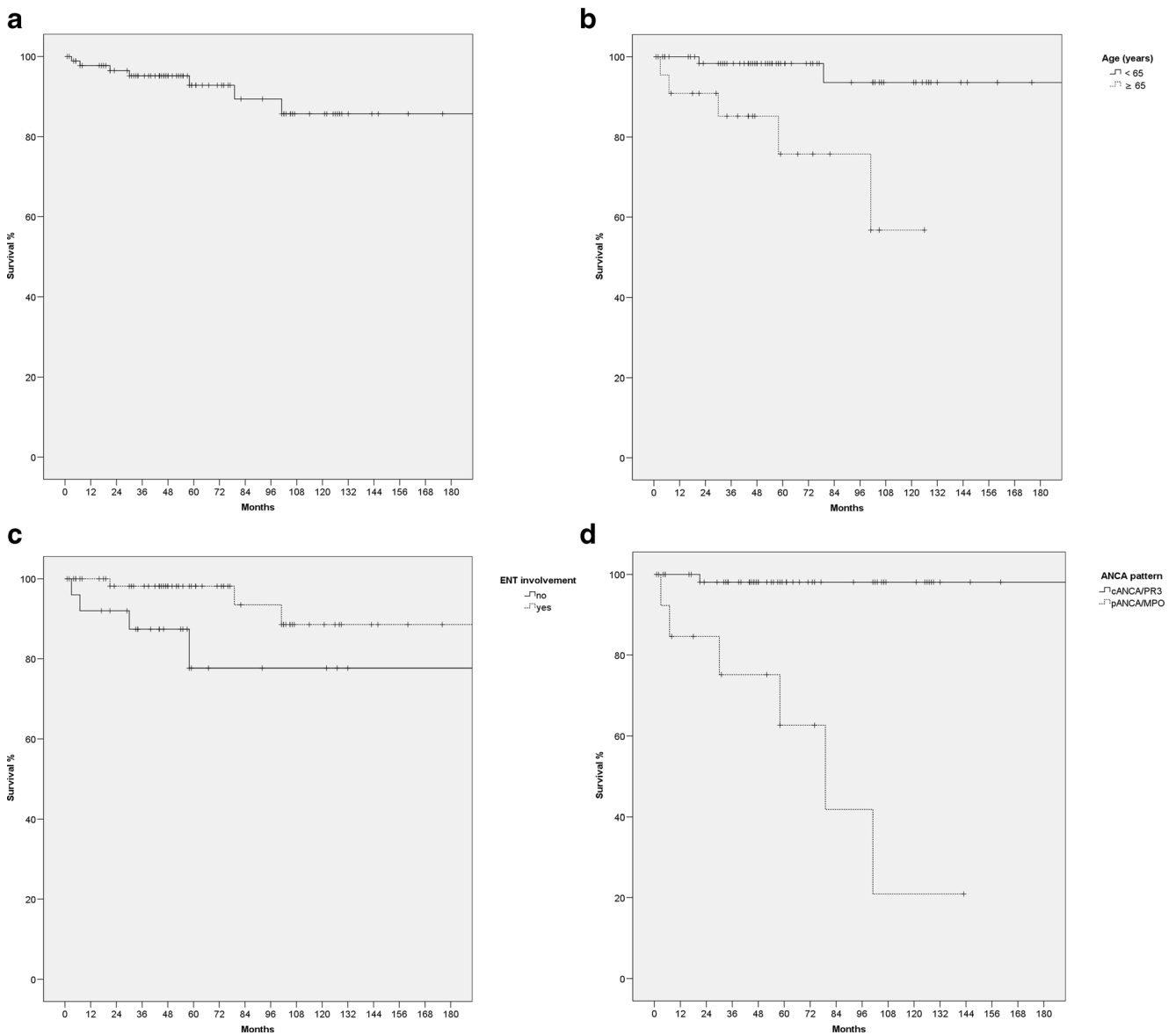


Fig. 3 Plots of survival rates during the follow-up. **a** Overall survival. **b** Survival rates in GPA patients according to the age, below and above 65 years of age. **c** Survival rates in GPA patients with and without ENT involvement. **d** Survival rates in GPA patients according to the ANCA pattern

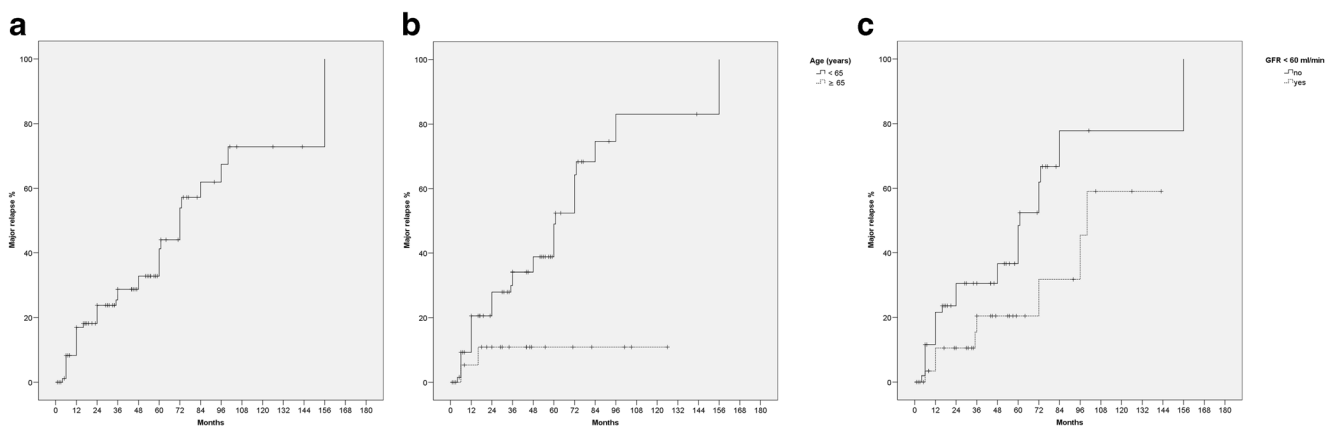


Fig. 4 Plots of relapse rates during follow-up. **a** Relapse rates in GPA patients according to the age, below and above 65 years of age. **c** Relapse rates in GPA patients according to renal impairment

Moreover, non-ENT-GPA patients had a greater probability of having a renal involvement [21 (84%) vs 35 (54.7%)], with a more severe disease. Our findings reporting lower levels of serum creatinine and lower rate of AKI and higher levels of eGFR in the ENT-GPA group confirmed what was observed by Rahmattulla and colleagues in 2015, who showed a higher eGFR at diagnosis and at 5-year follow-up in ANCA-associated vasculitis patients with ENT manifestation [21].

Generally, it seems that an ENT involvement is associated with milder forms of GPA as suggested by studies reporting an increased mortality risk in non-ENT-GPA patients [30–32].

Interestingly, ANCA status did not differ between the two groups, in contrast to previously reported data, in which ENT involvement is more frequently observed in anti-PR3 patients, while MPO specificity is significantly higher in patients with renal involvement [33]. This could reflect the particular composition of our cohort, composed by only “real-life” GPA patients, while the main literature data on ANCA significance on clinical subset are derived from RCT cohorts. It was already demonstrated that RCT patients are more frequently classified as MPA and presented worse prognostic features than the observational cohort (older age, higher BVASv3 and increased renal involvement) [34].

One of the limits of our study is the low prevalence of GPA-MPO subset (13 patients) within our court, which may not allow detecting a real clinical difference between the groups. Indeed, it has been recently demonstrated that GPA-MPO patients differ from the classic MPO-positive phenotype because they frequently present with limited or mild disease, with high rate of ENT symptoms and less renal disease [13, 14].

This suggests that the ANCA antibody criterion may not be enough to fully describe the wide spectrum of clinical subsets in GPA, while it has been recently reported that serological classification could predict relapses, but not survival or renal outcome in renal biopsy-proven ANCA-associated vasculitis [35]. Moreover, in the cluster analysis, Mahr et al. considered patients without renal involvement (mainly with ENT symptoms) into a separate group with better survival, irrespectively of ANCA type [36].

The low overall mortality observed, if compared to other RCTs [37], may be caused by a higher prevalence in our series of outpatients, thus not in the acute phase of the disease. However, in the last decade, we are witnessing an improvement survival in GPA and our results may reflect this recent trend [38]. Furthermore, our ENT-GPA patients showed a lower mortality if compared to non-ENT-GPA, but it has to be considered that they are younger.

The exiguity of our cohort, if compared to other series, and the retrospective design of our study are additional limits. On the contrary, this is the first Italian report on the ENT manifestations among GPA patients.

Conclusions

ENT involvement, as an early sign of GPA disease, yields the otorhinolaryngologist a favoured position for the diagnosis. A tight collaboration between ENT surgeons and rheumatologists is a crucial element for an early diagnosis and treatment. This is particularly true in those ANCA-negative forms whose diagnosis is often challenging and delayed.

Our data confirm that ENT involvement is not only one of the key clinical features of GPA, but also point out a milder disease subset with lower renal involvement and could predict a lower mortality rate.

Compliance with ethical standards

Disclosures None.

References

1. Falk RJ, Gross WL, Guillevin L, Hoffman GS, Jayne DRW, Jennette JC, Kallenberg CGM, Luqmani R, Mahr AD, Matteson EL, Merkel PA, Specks U, Watts RA, American College of Rheumatology, American Society of Nephrology, European League Against Rheumatism (2011) Granulomatosis with polyangiitis (Wegener's): an alternative name for Wegener's granulomatosis. *Arthritis Rheum* 63(4):863–864. <https://doi.org/10.1002/art.30286>
2. Holle JU, Laudien M, Gross WL (2010) Clinical manifestations and treatment of Wegener's granulomatosis. *Rheum Dis Clin N Am* 36(3):507–526. <https://doi.org/10.1016/j.rdc.2010.05.008>
3. Cote MF, Hoffman GS, Yerg DE, Kaufman GI, Targonski P, Kaslow RA (1996) The epidemiology of Wegener's granulomatosis. Estimates of the five-year period prevalence, annual mortality, and geographic disease distribution from population-based data sources. *Arthritis Rheum* 39(1):87–92. <https://doi.org/10.1002/art.1780390112>
4. Mohammad AJ, Jacobsson LTH, Mahr AD, Sturfelt G, Segelmark M (2007) Prevalence of Wegener's granulomatosis, microscopic polyangiitis, polyarteritis nodosa and Churg-Strauss syndrome within a defined population in southern Sweden. *Rheumatology (Oxford)* 46(8):1329–1337. <https://doi.org/10.1093/rheumatology/kem107>
5. Mukhtyar C, Guillevin L, Cid MC, Dasgupta B, de Groot K, Gross W, Hauser T, Hellmich B, Jayne D, Kallenberg CGM, Merkel PA, Raspe H, Salvarani C, Scott DGI, Stegeman C, Watts R, Westman K, Witter J, Yazici H, Luqmani R, for the European Vasculitis Study Group (2009) EULAR recommendations for the management of primary small and medium vessel vasculitis. *Ann Rheum Dis* 68(3):310–317. <https://doi.org/10.1136/ard.2008.088096>
6. Tsuzuki K, Fukazawa K, Takebayashi H, Hashimoto K, Sakagami M (2009) Difficulty of diagnosing Wegener's granulomatosis in the head and neck region. *Auris Nasus Larynx* 36(1):64–70. <https://doi.org/10.1016/j.anl.2008.02.003>
7. Garcia-Porrúa C, Amor-Dorado JC, Gonzalez-Gay MA (2001) Unilateral submandibular swelling as unique presentation of Wegener's granulomatosis. *Rheumatology (Oxford)* 40(8):953–954. <https://doi.org/10.1093/rheumatology/40.8.953>
8. Wiik A (2002) Rational use of ANCA in the diagnosis of vasculitis. *Rheumatology (Oxford)* 41(5):481–483. <https://doi.org/10.1093/rheumatology/41.5.481>

9. Jennings CR, Jones NS, Dugar J, Powell RJ, Lowe J (1998) Wegener's granulomatosis—a review of diagnosis and treatment in 53 subjects. *Rhinology* 36(4):188–191
10. McDonald TJ, DeRemee RA (1993) Head and neck involvement in Wegener's granulomatosis (WG). *Adv Exp Med Biol* 336:309–313. https://doi.org/10.1007/978-1-4757-9182-2_49
11. Martinez Del Pero M, Rasmussen N, Chaudhry A, Jani P, Jayne D (2013) Structured clinical assessment of the ear, nose and throat in patients with granulomatosis with polyangiitis (Wegener's). *Eur Arch Otorhinolaryngol* 270(1):345–354. <https://doi.org/10.1007/s00405-012-2110-8>
12. Hernández-Rodríguez J, Hoffman GS, Koenig CL (2010) Surgical interventions and local therapy for Wegener's granulomatosis. *Curr Opin Rheumatol* 22(1):29–36. <https://doi.org/10.1097/BOR.0b013e328333e9e9>
13. Ono N, Niino H, Ueda A, Sawabe T, Nishizaka H, Furugo I, Yoshizawa S, Yoshizawa S, Tsukamoto H, Kiyohara C, Tada Y, Horiuchi T (2015) Characteristics of MPO-ANCA-positive granulomatosis with polyangiitis: a retrospective multi-center study in Japan. *Rheumatol Int* 35(3):555–559. <https://doi.org/10.1007/s00296-014-3106-z>
14. Schirmer JH, Wright MN, Herrmann K et al (2016) Myeloperoxidase-antineutrophil cytoplasmic antibody (ANCA)-positive granulomatosis with polyangiitis (Wegener's) is a clinically distinct subset of ANCA-associated vasculitis: a retrospective analysis of 315 patients from a German vasculitis referral center. *Arthritis Rheumatol* (Hoboken, N.J.) 68(12):2953–2963
15. Watts R, Lane S, Hanslik T, Hauser T, Hellmich B, Koldingsnes W, Mahr A, Segelmark M, Cohen-Tervaert JW, Scott D (2007) Development and validation of a consensus methodology for the classification of the ANCA-associated vasculitides and polyarteritis nodosa for epidemiological studies. *Ann Rheum Dis* 66(2):222–227. <https://doi.org/10.1136/ard.2006.054593>
16. Mukhtyar C, Lee R, Brown D, Carruthers D, Dasgupta B, Dubey S, Flossmann O, Hall C, Hollywood J, Jayne D, Jones R, Lanyon P, Muir A, Scott D, Young L, Luqmani RA (2009) Modification and validation of the Birmingham Vasculitis Activity Score (version 3). *Ann Rheum Dis* 68(12):1827–1832. <https://doi.org/10.1136/ard.2008.101279>
17. Exley AR, Bacon PA, Luqmani RA, Kitas GD, Gordon C, Savage COS, Adu D (1997) Development and initial validation of the Vasculitis Damage Index for the standardized clinical assessment of damage in the systemic vasculitides. *Arthritis Rheum* 40(2):371–380. <https://doi.org/10.1002/art.1780400222>
18. Levey AS, Stevens LA, Schmid CH, Zhang Y(L), Castro AF III, Feldman HI, Kusek JW, Eggers P, van Lente F, Greene T, Coresh J, for the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) (2009) A new equation to estimate glomerular filtration rate. *Ann Intern Med* 150(9):604–612. <https://doi.org/10.7326/0003-4819-150-9-200905050-00006>
19. Bellomo R, Ronco C, Kellum JA, Mehta RL, Palevsky P, Acute Dialysis Quality Initiative workgroup (2004) Acute renal failure—definition, outcome measures, animal models, fluid therapy and information technology needs: the Second International Consensus Conference of the Acute Dialysis Quality Initiative (ADQI) Group. *Crit Care* 8(4):R204
20. Hauer HA, Bajema IM, van Houwelingen HC et al (2002) Renal histology in ANCA-associated vasculitis: differences between diagnostic and serologic subgroups. *Kidney Int* 61(1):80–89. <https://doi.org/10.1046/j.1523-1755.2002.00089.x>
21. Rahmattulla C, de Lind van Wijngarden RA, Berden AE et al (2015) Renal function and ear, nose, throat involvement in antineutrophil cytoplasmic antibody-associated vasculitis: prospective data from the European Vasculitis Society clinical trials. *Rheumatology (Oxford)* 54(5):899–907. <https://doi.org/10.1093/rheumatology/keu357>
22. Hoffman GS, Kerr GS, Leavitt RY, Hallahan CW, Lebovics RS, Travis WD, Rottem M, Fauci AS (1992) Wegener granulomatosis: an analysis of 158 patients. *Ann Intern Med* 116(6):488–498. <https://doi.org/10.7326/0003-4819-116-6-488>
23. Sproson EL, Jones NS, Al-Deiri M, Lanyon P (2007) Lessons learnt in the management of Wegener's granulomatosis: long-term follow-up of 60 patients. *Rhinology* 45(1):63–67
24. McDonald TJ, DeRemee RA (1983) Wegener's granulomatosis. *Laryngoscope* 93(2):220–231
25. Wojciechowska J, Krajewski W, Krajewski P, Krecicki T (2016) Granulomatosis with polyangiitis in otolaryngologist practice: a review of current knowledge. *Clin Exp Otorhinolaryngol* 9(1):8–13. <https://doi.org/10.21053/ceo.2016.9.1.8>
26. Rahne T, Clauß F, Plontke SK, Keyßer G (2017) Prevalence of hearing impairment in patients with rheumatoid arthritis, granulomatosis with polyangiitis (GPA, Wegener's granulomatosis), or systemic lupus erythematosus. *Clin Rheumatol* 36(7):1501–1510. <https://doi.org/10.1007/s10067-017-3651-4>
27. Gottschlich S, Ambrosch P, Kramkowski D, Laudien M, Buchelt T, Gross WL, Hellmich B (2006) Head and neck manifestations of Wegener's granulomatosis. *Rhinology* 44(4):227–233
28. Kornblut AD, Wolff SM, deFries HO, Fauci AS (1980) Wegener's granulomatosis. *Laryngoscope* 90(9):1453–1465. <https://doi.org/10.1288/00005537-198009000-00005>
29. Langford CA, Sneller MC, Hallahan CW, Hoffman GS, Kammerer WA, Talar-Williams C, Fauci AS, Lebovics RS (1996) Clinical features and therapeutic management of subglottic stenosis in patients with Wegener's granulomatosis. *Arthritis Rheum* 39(10):1754–1760. <https://doi.org/10.1002/art.1780391020>
30. Pavone L, Grasselli C, Chierici E, Maggione U, Garini G, Ronda N, Manganelli P, Pesci A, Rioda WT, Tumiatì B, Pavesi G, Vaglio A, Buzio C, Secondary and Primar Vasculitides (Se.Pri.Va) Study Group (2006) Outcome and prognostic factors during the course of primary small-vessel vasculitides. *J Rheumatol* 33(7):1299–1306
31. Bligny D, Mahr A, Le Toumelin P, Mouthon L, Guillevin L (2004) Predicting mortality in systemic Wegener's granulomatosis: a survival analysis based on 93 patients. *Arthritis Rheum* 51(1):83–91. <https://doi.org/10.1002/art.20082>
32. Mohammad AJ, Jacobsson LTH, Westman KWA, Sturfelt G, Segelmark M (2009) Incidence and survival rates in Wegener's granulomatosis, microscopic polyangiitis, Churg-Strauss syndrome and polyarteritis nodosa. *Rheumatology (Oxford)* 48(12):1560–1565. <https://doi.org/10.1093/rheumatology/kep304>
33. Lionaki S, Blyth ER, Hogan SL, Hu Y, Senior BA, Jennette CE, Nachman PH, Jennette JC, Falk RJ (2012) Classification of antineutrophil cytoplasmic autoantibody vasculitides: the role of antineutrophil cytoplasmic autoantibody specificity for myeloperoxidase or proteinase 3 in disease recognition and prognosis. *Arthritis Rheum* 64(10):3452–3462. <https://doi.org/10.1002/art.34562>
34. Pagnoux C, Carette S, Khalidi NA et al (2015) Comparability of patients with ANCA-associated vasculitis enrolled in clinical trials or in observational cohorts. *Clin Exp Rheumatol* 33(2 Suppl 89):S77–S83
35. Córdova-Sánchez BM, Mejía-Vilet JM, Morales-Buenrostro LE, Loyola-Rodríguez G, Uribe-Uribe NO, Correa-Rotter R (2016) Clinical presentation and outcome prediction of clinical, serological, and histopathological classification schemes in ANCA-associated vasculitis with renal involvement. *Clin Rheumatol* 35(7):1805–1816. <https://doi.org/10.1007/s10067-016-3195-z>
36. Mahr A, Katsahian S, Varet H, Guillevin L, Hagen EC, Höglund P, Merkel PA, Pagnoux C, Rasmussen N, Westman K, Jayne DRW, for the French Vasculitis Study Group (FVSG) and the European Vasculitis Society (EUVAS) (2013) Revisiting the classification of clinical phenotypes of anti-neutrophil cytoplasmic antibody-

- associated vasculitis: a cluster analysis. *Ann Rheum Dis* 72(6): 1003–1010. <https://doi.org/10.1136/annrheumdis-2012-201750>
37. Flossmann O, Berden A, de Groot K, Hagen C, Harper L, Heijl C, Hoglund P, Jayne D, Luqmani R, Mahr A, Mukhtyar C, Pusey C, Rasmussen N, Stegeman C, Walsh M, Westman K, for the European Vasculitis Study Group (2011) Long-term patient survival in ANCA-associated vasculitis. *Ann Rheum Dis* 70(3): 488–494. <https://doi.org/10.1136/ard.2010.137778>
38. Wallace ZS, Lu N, Unizony S, Stone JH, Choi HK (2016) Improved survival in granulomatosis with polyangiitis: a general population-based study. *Semin Arthritis Rheum* 45(4):483–489. <https://doi.org/10.1016/j.semarthrit.2015.07.009>